

Eli Boyden

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EDUCATION

University of Virginia, School of Engineering and Applied Science

August 2023 – May 2027

Bachelor of Science in Computer Science, Data Science Minor

- Relevant Coursework: Data Structures, Algorithms, Software Development Essentials, Computer Systems Architecture, Software Engineering, Cybersecurity, Computer Networks

Bachelor of Arts in Mathematics, Concentration in Probability and Statistics

- Relevant Coursework: Probability, Linear Algebra, Differential Equations, Stochastic Processes, Machine Learning

Cumulative GPA: 3.954/4.00

EXPERIENCE

Software Engineer Intern

August 2025 – December 2025

City of Charlottesville Department of IT

Charlottesville, VA

- Integrating PostgreSQL cloud database with Supabase, creating a RESTful API for storage and manipulation of alert updates and user account authentication data.
- Accelerated cloud adoption by automating collection, processing and analysis of API and RSS feed data on critical events in Charlottesville using specialized Python AWS Lambda serverless functions.

Full Stack Software Engineer Intern

May 2025 – August 2025

YSTEM and Chess

Remote

- Wrote and maintained automated unit and integration tests using Jest, reducing React front-end production bugs and increasing development team test coverage by over 60%.
- Lead accessibility improvements across React components, increasing compliance with WCAG standards, and improving usability for screen readers, keyboard navigation, and users with visual impairments.

Application Software Developer

May 2025

Mathesis Technology

Hybrid Charlottesville, VA

- Launched secure departmental JavaScript, HTML, CSS course sites on Cloudflare's global network improving access while reducing manual workload for faculty.
- Adapted, deployed and maintained TypeScript static site generator serving 150+ students per semester.
- Collaborating with faculty to identify website needs and tailoring technical solutions accordingly, ensuring consistent functionality and aesthetic presentation across all devices and browsers.

PROJECTS

TAGuide | PostgreSQL, Python, Encryption (bcrypt, Fernet), Flask, React

- Full-stack encrypted React application for running Selenium web scraping jobs to automate data collection from Gradescope, enabling tracking and visualization of Teaching Assistant (TA) grading progress.
- Helped identify and subsequently correct grading work imbalances of up to 25%.

Artificial Neural Network | Java, JUnit, Git

- Achieved 97% accuracy on MNIST handwritten digit dataset classification and 91% on heart attack prediction by optimizing matrix operations and refining model architecture.
- Validated and tested implementations with custom unit tests using JUnit, ensuring model stability and correctness through structured experimentation.

SKILLS

Programming Languages: Python, Java, JavaScript, TypeScript, Go, C, SQL, HTML, CSS, SCSS (Sass)

Frameworks & Libraries: React.js, Node.js, Express.js, Flask, Jest, JUnit, JPA, Selenium, Gradle, Django

Tools & Platforms: Git, AWS, Cloudflare Pages/Workers, Cloudflare Access, Make

Concepts & Methodologies: Object-Oriented Programming (OOP), Agile Development, Scrum, Unit Testing, Accessibility Compliance, RESTful APIs, CI/CD Basics, Machine Learning (Neural Networks)

Databases: PostgreSQL, MongoDB, SQLite, Supabase

CERTIFICATIONS

AWS: Certified Cloud Practitioner CLF-C02 | [Certification](#)

Scrimba: Advanced React | [Certification](#)